Noah Pettinari

he/him/his

+39 352 213 1150 | noah.pettinari@gmail.com

Education

Master of Science in Astrophysics and Cosmology — **September 2026 - Present**
Università degli Studi di Padova — **Padua, PD, Italy**
Laurea Magistrale

Bachelor of Science in Physics — **June 2022**
University of Oregon — **Eugene, OR, USA**
Minors in Biology and Mathematics
Summa Cum Laude, Departmental Honors

Professional Experience

Senior Operations Specialist — **April 2025 - August 2026**
Tandem, Inc. — **New York, NY, USA**

- Efficiently utilized custom AI prediction to inform healthcare providers of predicted prior authorization denials with >90% accuracy.
- Wrote appeals for healthcare insurance denials of prior authorizations to advocate for patient access to healthcare and ensure timely access to prescribed medication.
- Assisted with onboarding, training, and ramping-up of new employees to ensure HIPAA compliance and timely submission of appeals for health insurance denials.

Computational Associate II — **August 2022 - August 2024**
The Broad Institute of MIT and Harvard, Nehme and McCarroll Labs — **Boston, MA, USA**

- Independently developed computational pipelines for processing and analyzing single-cell genetic perturbation RNA sequencing data in stem cell-derived brain cells.
- Analyzed high-throughput single-cell RNA sequencing data from 200+ single-cell transcriptomic libraries to elucidate the genetic basis for schizophrenia and drug response.
- Implemented a standardized pipeline for processing and analyzing high-throughput bulk RNA sequencing data to identify effects of various drugs at varying dosages and timed exposures.
- Performed latent factor analyses using machine learning methods to identify coordinated gene expression programs induced by various drug and genetic perturbations in an array of stem cell-derived and primary tissue cell types.
- Collaboratively developed an experimental framework to identify correlations between genotypic variation and drug response to antipsychotics and inflammatory proteins.
- Validated cell line identities of over 1,000 cell lines via periodic whole genome analysis to ensure systematic robustness of cell cultures and minimize experimental variability.
- Created, implemented, and utilized standardized training procedures for computational associates to improve the onboarding process and ensure quick learning of lab compliance standards.

Biophysics Research Assistant — **June 2019 - June 2022**
UO Physics Department, Parthasarathy Lab — **Eugene, OR, USA**

- Performed image analysis using Python and MATLAB to develop 3 image-based assays for quantifying zebrafish gut microbial interactions and interspecies competition dynamics.

- Designed and constructed 2 microbial assays and 1 light sheet fluorescence microscope for imaging fluorescent microbes on microscopic and macroscopic scales to validate existing biophysical theories.
- Collaborated with 2 independent research groups on interdisciplinary projects that led to 3 presentations at university conferences and 1 publication in a world renowned journal.
- Conducted experiments with BSL2+ microbes associated with vertebrates to elucidate metabolic species interactions and promote DNA engineering, cultivation, and editing of 6+ bacterial strains.
- Trained incoming undergraduate students by independently developing new SOPs for standard lab experiments such as fluorescence microscopy, standard image analysis, and bacterial cell culturing.

RISE Physics Research Intern **July 2021 - September 2021**
WWU-Münster, Nonlinear Photonics Group **Münster, Germany**

- Developed and constructed a two-photon polymerization via direct laser writing microscope for engineering of biocompatible cell assays using a near-infrared femtosecond pulse laser.
- Established 5+ protocols for biocompatible polymer gels that allowed for measurement of mechanical properties of yeast cells, enzymes, and hydrogel compatibility testing.
- Designed and troubleshooted a custom-built GUI interface of microscope hardware products to streamline and simplify data acquisition.
- Tested polymerization efficiency on polymer hydrogels to troubleshoot and repair microscope faults.

Library Student Assistant III **January 2019 - June 2022**
University of Oregon Libraries **Eugene, OR, USA**

- Implemented and organized 3+ regular workshops on LaTeX for peers and library patrons.
- Tutored library patrons on math homework on topics ranging from elementary algebra to multivariable calculus.
- Forged relationships with library donors to continue and increase financial support of library spaces for an additional 3 years, which led to a 10% increase in library resource availability and usage.

Skills

Programming languages: Python (numpy, matplotlib, scipy, pandas), R, Bash, Unix, Git, LaTeX, MATLAB, ImageJ, Fiji
Computational analysis: High-performance computing, parallel computing, image analysis, multivariate statistical analysis, machine learning, dimensional reduction, Github, physical modeling and simulations, stochastic simulations (Monte Carlo, Markov chains, etc), numerical modeling, latent factor analysis, Arduino
Qualitative: Experimental project design, communicative team player with experience managing several experimental groups at a time, troubleshooting, creative problem solving, highly motivated, science communication, independently driven

Publications

Rachel A Battaglia; Sonia Bolshakova; Ilinca Mazureac; Dhara Liyanage; **Noah Pettinari**; Autumn Johnson; Ethan Crouse; Sartaj Habib; Isabel Flessas; Ajay Nadig; Derek Hawes; Matthew Tegtmeyer; Caroline Becker; Sulagna Ghosh; Giulio Genovese; Marina Hogan; Adrianna Maglieri; Lindy E Barrett; Laurence Daheron; Steven A McCarroll; Ralda Nehme. “Gene editing in "cell villages" enables exploring disease-relevant mutations in many genetic backgrounds.” bioRxiv 2025.11.08.687374 [Preprint] doi: https://doi.org/10.1101/2025.11.08.687374

Milena M Andzelm, **Noah Pettinari**, Sonia Bolshakova, Matthew Tegtmeyer, Dan Meyer, Autumn Johnson, Curtis J Mello, Connie T Yu, Ilinca Mazureac, Giulio Genovese, Adrianna Maglieri, Kiku Ichihara, Marina Hogan, Derek Hawes, Steven A McCarroll, Ralda Nehme. "Human genetic variation shapes the response of neurons to interferons." bioRxiv 2025.05.28.653507 [Preprint]. doi: https://doi.org/10.1101/2025.05.28.653507

Sundarraman, Deepika, Edouard A. Hay, Dylan M. Martins, Drew S. Shields, **Noah L. Pettinari**, and Raghuveer Parthasarathy. "Higher-Order Interactions Dampen Pairwise Competition in the Zebrafish Gut Microbiome." *Mbio* 11, no. 5 (October 2020): e01667-20. https://doi.org/10.1128/mBio.01667-20.

Presentations

Pettinari, Noah, Milena Andzelm, Jim Nemesh, Matthew Tegtmeyer, Adrianna Maglieri, Marina Hogan, Kiku Ichihara, Steve McCarroll, and Ralda Nehme (2024). Identifying sources of regulatory variation in the human neuronal response to interferons. Stanley Center for Psychiatric Research Scientific Advisory Board (SAB) Meeting 2024, Cambridge, MA. ***Poster***.

Pettinari, Noah (2024). Human Genetic Variation in the Neuronal Response to Interferons. Broad Institute Cellular Models Meeting, Cambridge, MA. ***Oral presentation***.

Pettinari, Noah, Milena Andzelm, Matthew Tegtmeyer, Jim Nemesh, Steve McCarroll, and Ralda Nehme (2023). Identifying context-dependent eQTLs in stem-cell derived neurons. Broad Institute Annual Retreat, Boston, MA. ***Poster***.

Pettinari, Noah, Matthew Tegtmeyer, Milena Andzelm, Dhara Liyanage, Jim Nemesh, Marina Hogan, Kiku Ichihara, Ralda Nehme, and Steve McCarroll (2023). Latent factor analysis of neural cell type responses to *in vitro* perturbations. Stanley Center for Psychiatric Research Scientific Advisory Board (SAB) Meeting 2023, Cambridge, MA. ***Poster***.

Pettinari, Noah and Raghuveer Parthasarathy (2021). Bacterial range expansion and the Fisher speed: a discrepancy in nutrient-rich media. University of Oregon Undergraduate Research Symposium, Eugene, OR. https://youtu.be/xfx2LYk63-A. ***Poster***.

Pettinari, Noah and Raghuveer Parthasarathy (2020). Characterizing the relationship between bacterial motility and range expansion. University of Oregon Undergraduate Research Symposium, Eugene, OR. https://youtu.be/HGtZQec32WY. ***Poster***.

Scholarships, Awards, and Grants

2022 Phi Beta Kappa Honors Society

2021 Audria M. Edwards Scholarship
DAAD RISE Intern
Clarence and Lucille Dunbar Scholarship
Mercer Family Foundation Scholarship
General University Scholarship

2020 Presidential Undergraduate Research Scholarship
Presidential Undergraduate Research Scholars Research Stipend
Thomas E. Mattson Scholarship
Grace Morris Scholarship
General University Scholarship

2019 Clarence and Lucille Dunbar Scholarship

2018 Scholarships for Oregon Scientists II
Audria M. Edwards Scholarship

Maria C. Jackson and General George A. White Scholarship
PathwayOregon Grant
Summit Scholarship

Organizations, Outreach, and Community Service

2023-2024 **Broad RATs United Union**, *Coordinating Committee*
Collaborated with fellow organizers and interfaced with United Electrical, Radio, and Machine Workers of America (UE) to initiate, plan, strategize, and organize a union of 500+ entry-level researchers at the Broad Institute of MIT and Harvard to advocate for better working conditions.

2023-2024 **Broad Institute Scientists in the Classroom Program**, *Volunteer*
Collaborated with Broad Institute coordinators and Cambridge public schools faculty to develop middle school biology lesson plans to promote engagement with fundamental biology topics including genetics and bioethics.

2023-2024 **Broad Institute Meet-a-Scientist Program**, *Volunteer*

2018-2022 **Society of Physics Students**, *Student Member*

2021-2022 **Out in STEM University of Oregon Chapter**, *Vice President of Finance*
Worked with club leadership to create and lead events related to professional development targeted towards LGBTQ+ students in STEM fields. Managed club finances, budget, and funding.

2019-2020 **Eugene Math Festival**, *Volunteer*
Access Network, *Network Fellow*

2019 **LaTeX Workshop**, *Organizer*
Co-organizer: Cameron Cunningham
Developed and led a curriculum for a one day workshop covering the basics of the LaTeX typesetting language.

2019 **The North Star Project**, *Volunteer*
Supervisor: Rachael Klaiss
Assisted with professional development and diversity activities for incoming first-year students in STEM with a focus on astronomy.